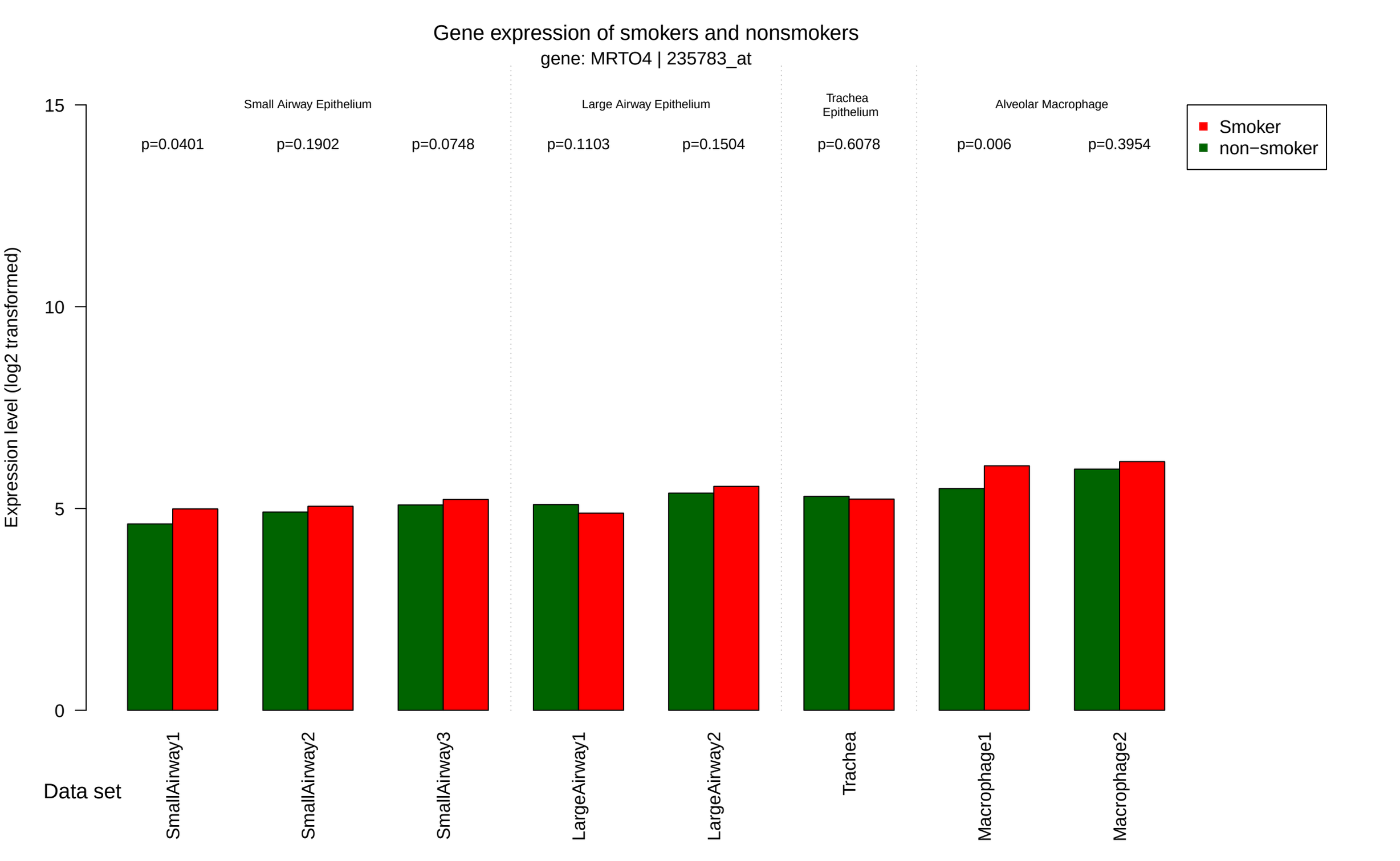
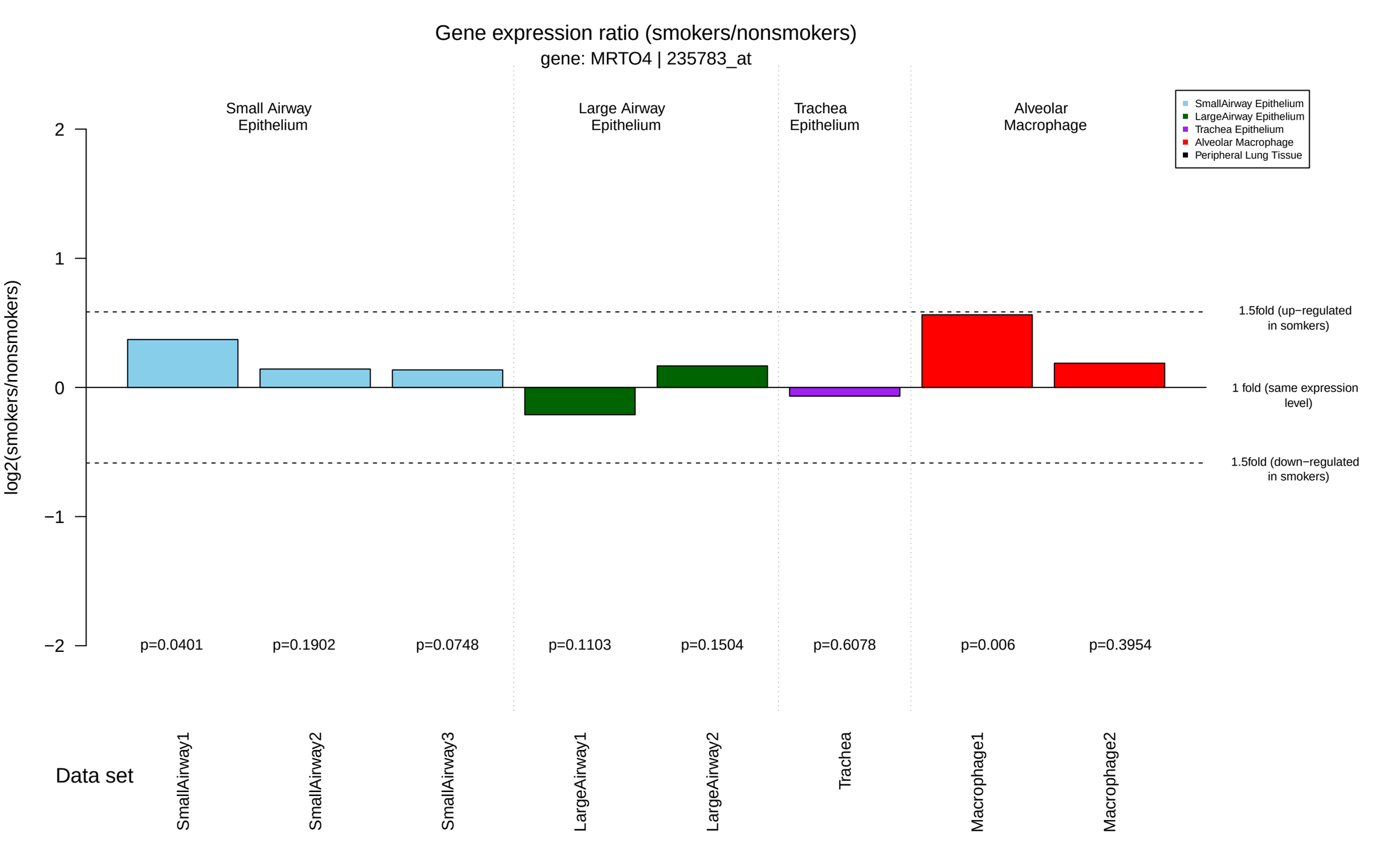
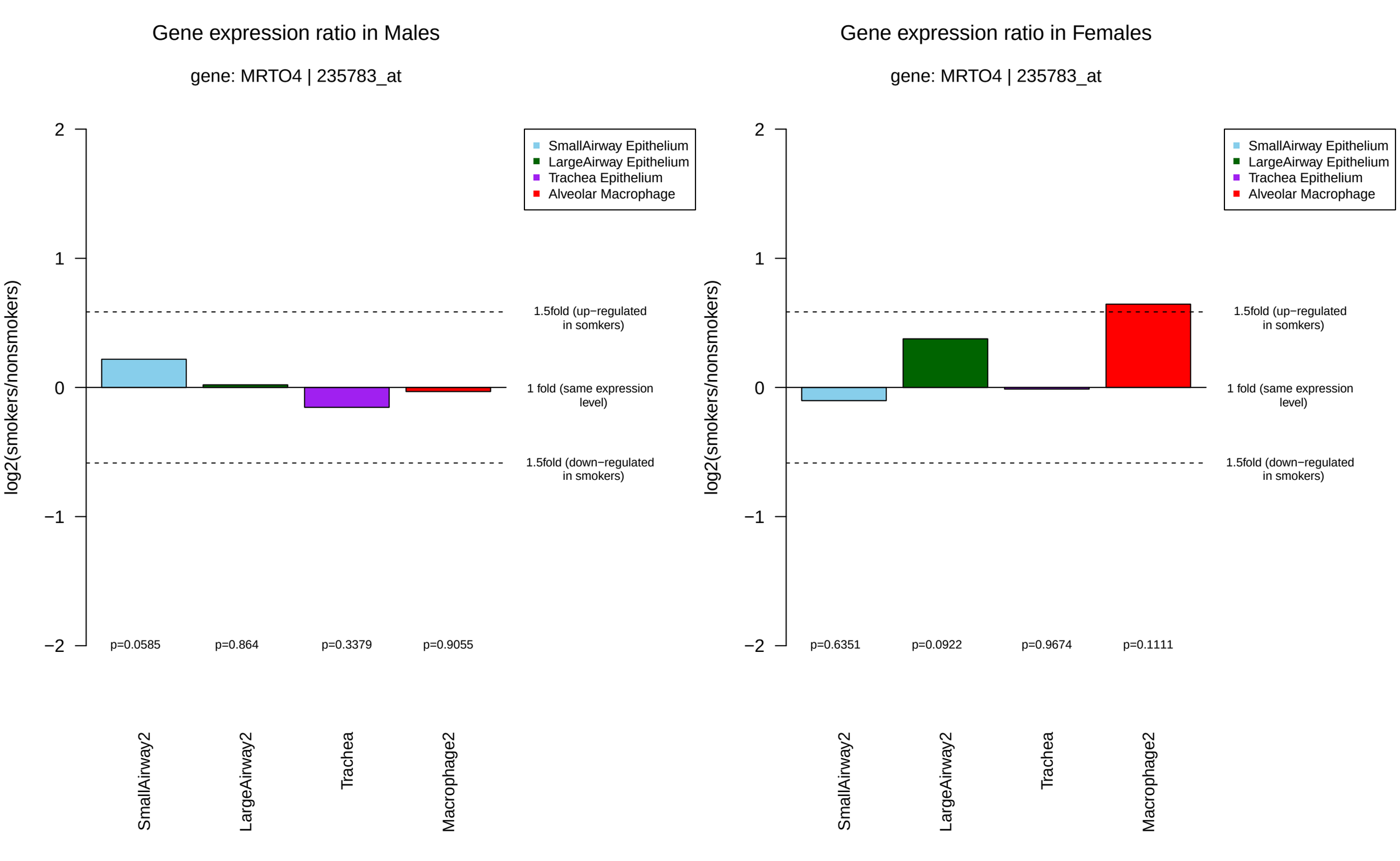
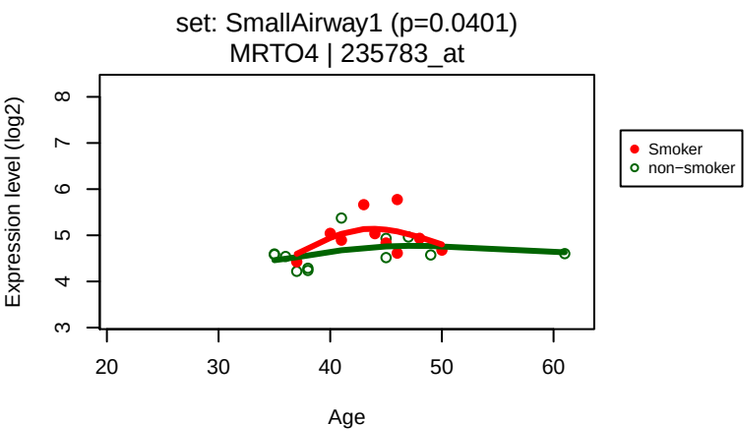
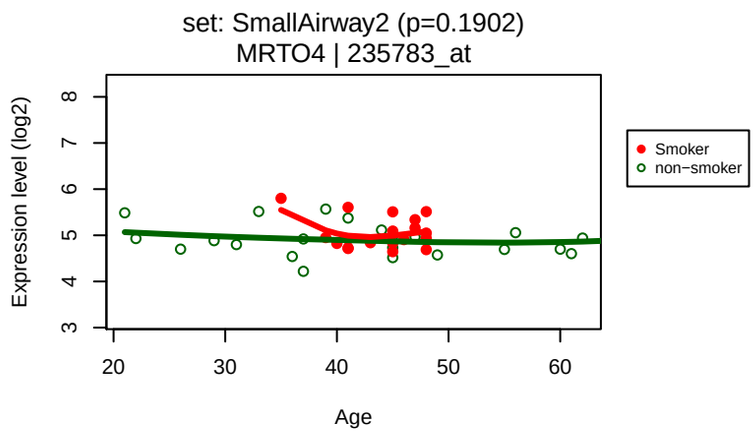
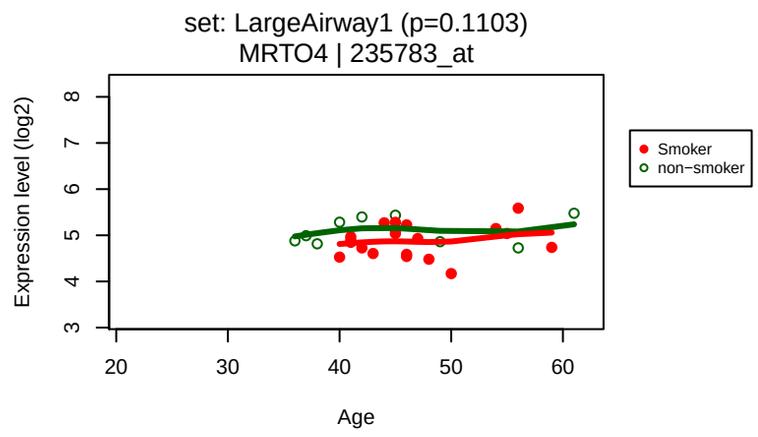

Gene expression of smokers and nonsmokers
gene: MRTO4 | 235783_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0401
p=0.1902
p=0.0748
p=0.1103
p=0.1504
p=0.6078
p=0.006
p=0.3954
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

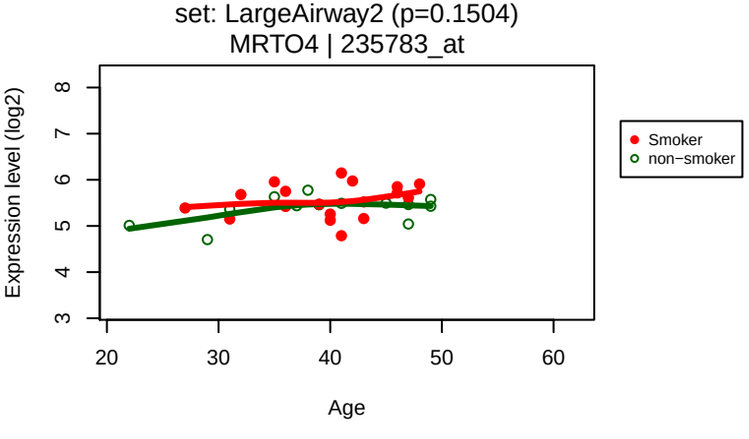

Gene expression ratio (smokers/nonsmokers)
gene: MRTO4 | 235783_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0401
p=0.1902
p=0.0748
p=0.1103
p=0.1504
p=0.6078
p=0.006
p=0.3954
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

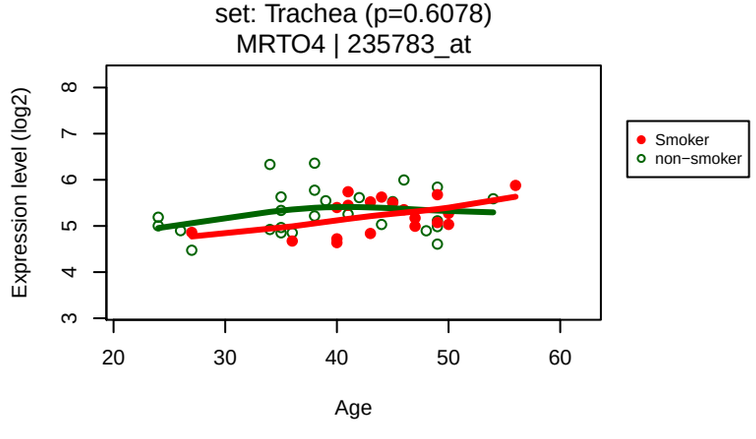

Gene expression ratio in Males
gene: MRTO4 | 235783_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0585
p=0.864
p=0.3379
p=0.9055
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRTO4 | 235783_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6351
p=0.0922
p=0.9674
p=0.1111
SmallAirway2
LargeAirway2
Trachea
Macrophage2

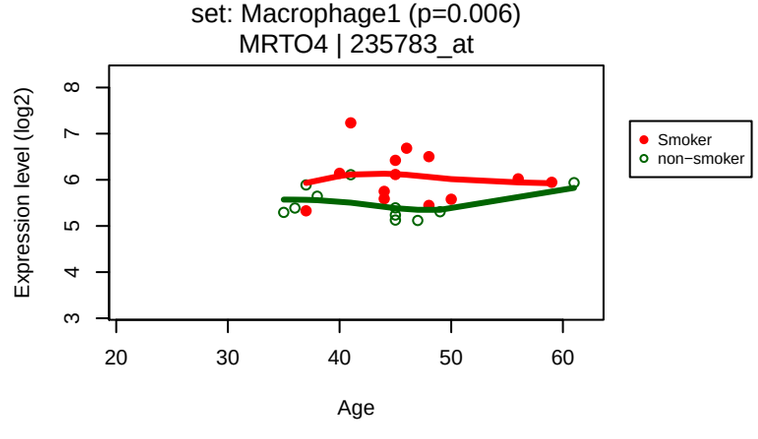

set: SmallAirway1 (p=0.0401)
MRTO4 | 235783_at
Expression level (log2)
Age
Smoker
non−smoker

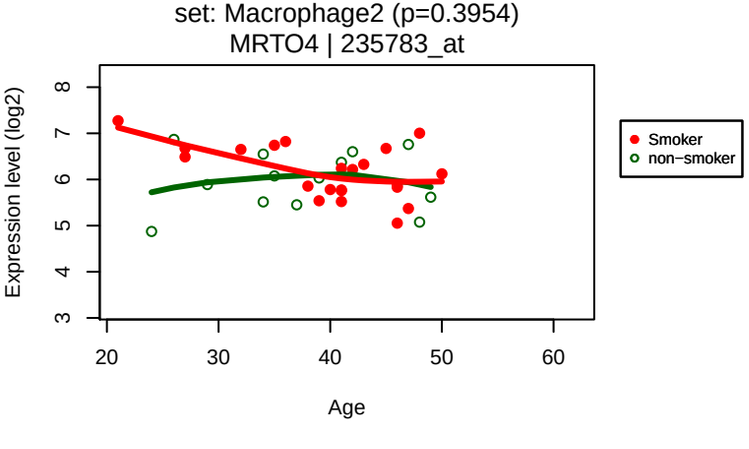

set: SmallAirway2 (p=0.1902)
MRTO4 | 235783_at
Expression level (log2)
Age
Smoker
non−smoker

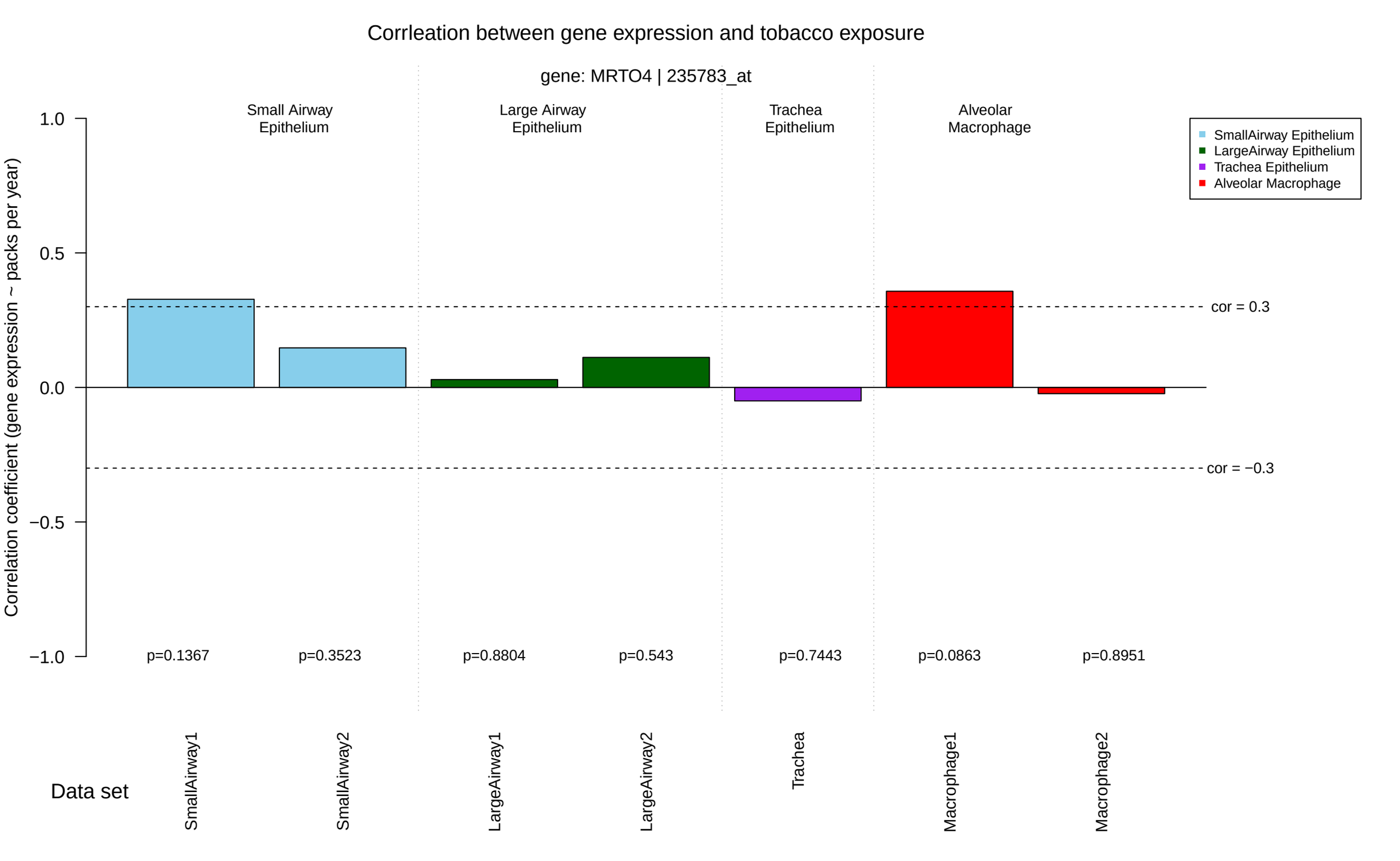

set: LargeAirway1 (p=0.1103)
MRTO4 | 235783_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1504)
MRTO4 | 235783_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6078)
MRTO4 | 235783_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.006)
MRTO4 | 235783_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3954)
MRTO4 | 235783_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRTO4 | 235783_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1367
p=0.3523
p=0.8804
p=0.543
p=0.7443
p=0.0863
p=0.8951
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2